



# **Setup a working website powered by Mediawiki on a VM running LAMP stack n a VM running LAMP stack**

Designed by-

Vijaylaxmi(293952)

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# Objective

To deploy a fully functional **MediaWiki** website on a **virtual machine** running the **LAMP stack (Linux, Apache, MySQL, PHP)**, and to implement a **daily backup mechanism** for both the MySQL database and MediaWiki file system to enable disaster recovery.

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# Introduction

**Task:** Deploy a working MediaWiki website on a VM running the LAMP stack (Linux, Apache, MySQL, PHP).

**SUB task:** Implement a robust daily backup system for MediaWiki data (database + uploaded files).

# What is LAMP?

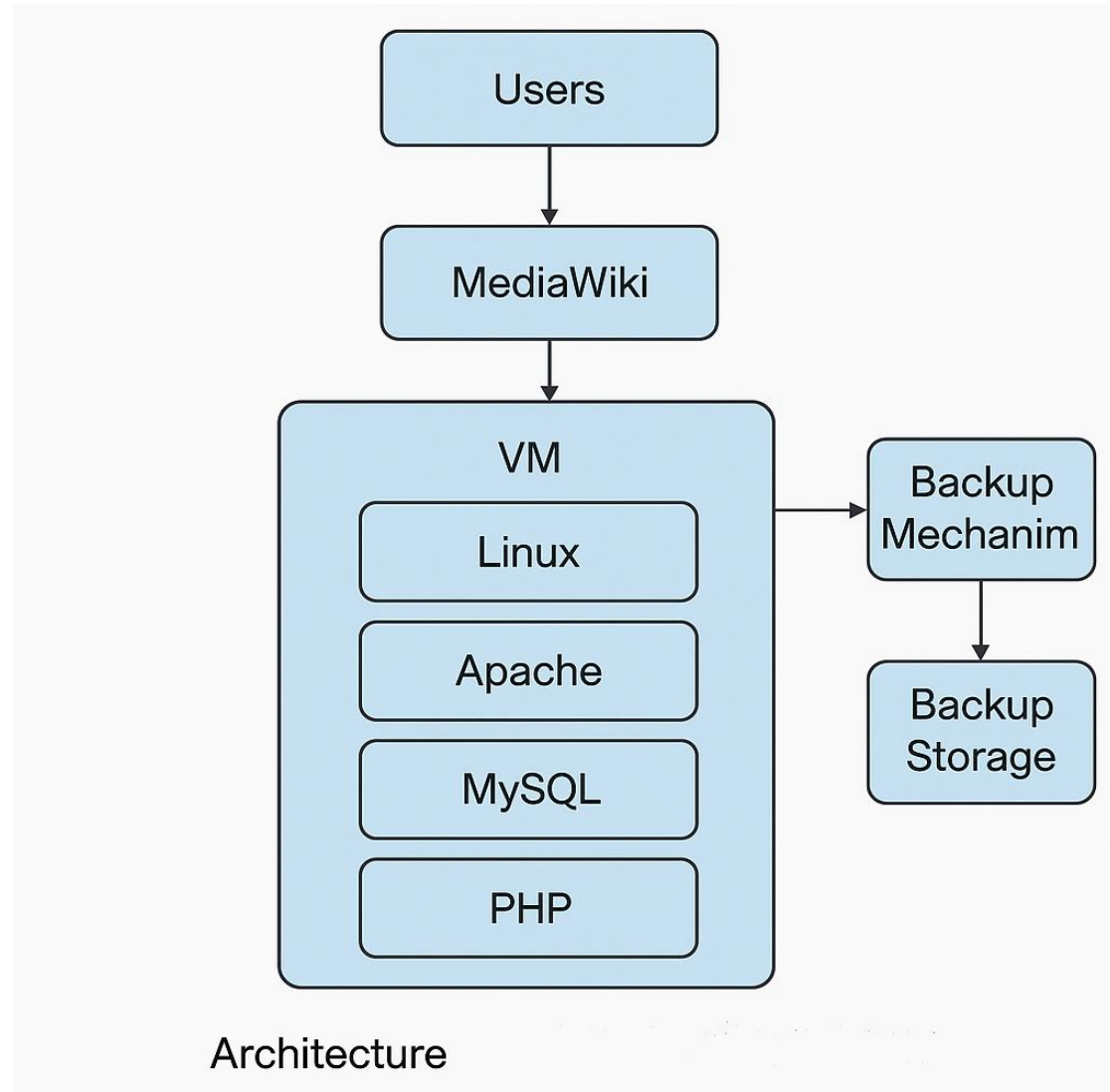
Linux –  
operating  
system

Apache -  
HTTP

Mysql -  
Database

Php -  
scripting

# Architecture diagram





# Virtual Machine Setup Overview

- A **virtual machine (VM)** is a software-based emulation of a physical computer. It allows you to run an entire operating system (like Ubuntu Linux) on top of your host OS (like Windows) using software such as **VirtualBox** or **VMware Workstation**. For this project, a VM provides a safe, isolated environment to deploy and test your MediaWiki website.
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# Hardware Requirements

## ➤ Host System:

- Processor: Intel i5 or better (64-bit architecture)
- RAM: Minimum 8 GB
- Disk: Minimum 30 GB free space
- OS: Windows/Linux/Mac (to install VirtualBox/VMWare)

## ➤ Virtual Machine:

- Guest OS: Ubuntu 22.04 LTS (recommended)
- CPU: 2 Cores
- RAM: 2 GB minimum
- Disk: 15–20 GB allocated storage

# Software Requirements

## ➤ **Host Software:**

- Oracle VirtualBox (or VMware)
- Ubuntu ISO file

## ➤ **Guest Software (within Ubuntu VM):**

- Apache2 (Web server)
- MySQL or MariaDB (Database)
- PHP and required extensions
- MediaWiki
- phpMyAdmin (optional, for DB GUI)



# Step-by-Step Installation Guide

## Step 1: Create and Set Up Virtual Machine

- Download VirtualBox from [virtualbox.org](https://www.virtualbox.org)
- Download Ubuntu Server ISO (22.04 LTS) from [ubuntu.com](https://ubuntu.com)
- Create a new VM in VirtualBox:
  - ✓ OS: Linux → Ubuntu (64-bit)
  - ✓ RAM: 2048 MB
  - ✓ Disk: 20 GB (dynamically allocated)
- Mount ISO and install Ubuntu
- During Ubuntu setup:
  - ✓ Set username/password
  - ✓ Choose minimal installation
  - ✓ Configure network or keep DHCP

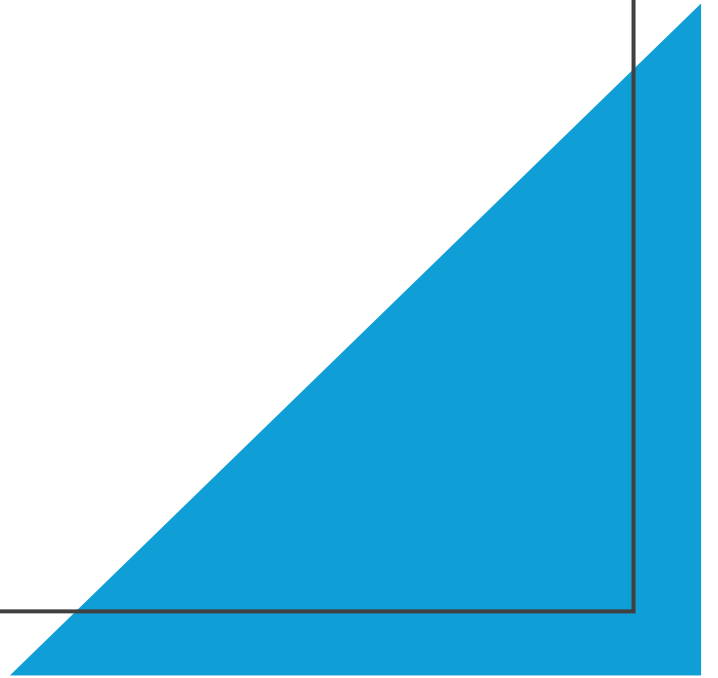
## **Step 2: Install Apache**

`sudo apt update`  
`sudo apt install apache2 -y`

`sudo systemctl enable apache2`

`sudo systemctl start apache`

Verify: `http://` shows Apache test page



### **Step 3: Install MySQL/MariaDB**

```
sudo apt install mysql-server -y sudo mysql_secure_installation
```

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```

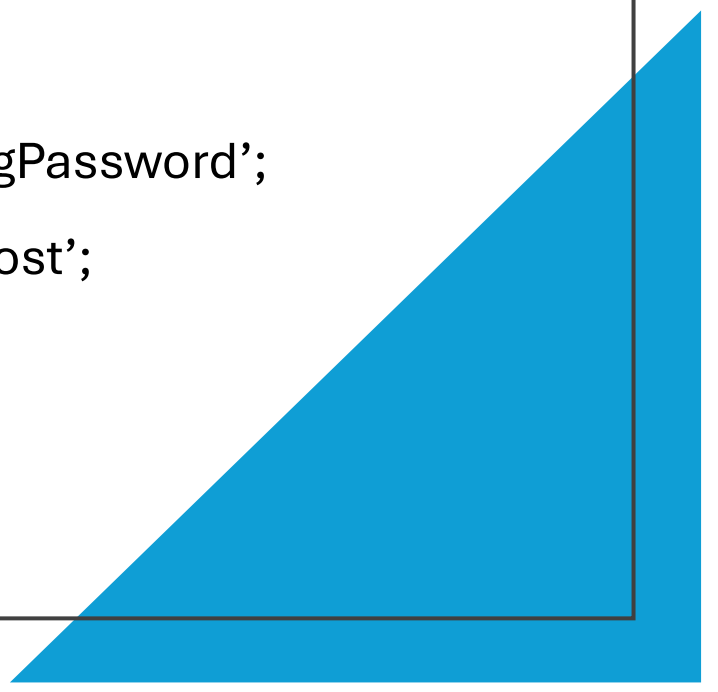
#### **Create database and user:**

```
CREATE DATABASE wikidb;
```

```
CREATE USER 'wikiuser'@'localhost' IDENTIFIED BY 'StrongPassword';
```

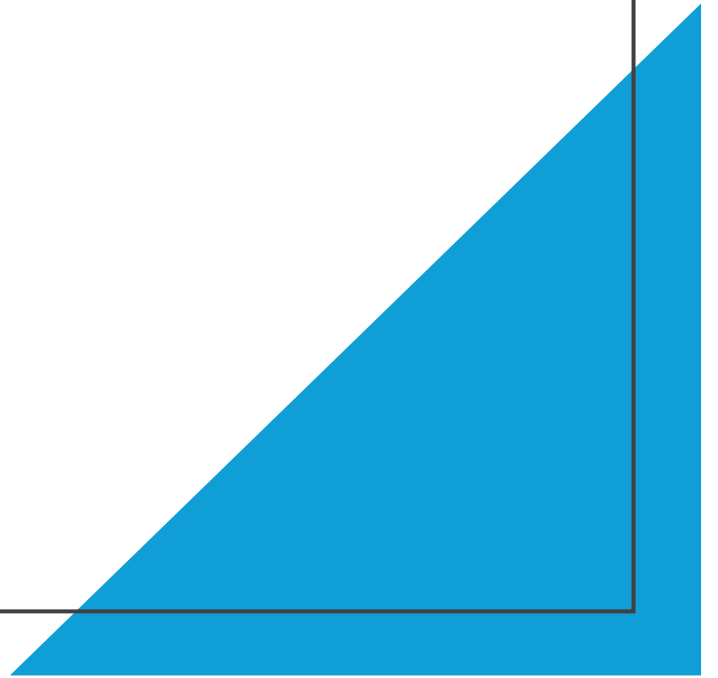
```
GRANT ALL PRIVILEGES ON wikidb.* TO 'wikiuser'@'localhost';
```

```
FLUSH PRIVILEGES;
```

A large blue right-angled triangle is positioned in the bottom right corner of the slide, with its hypotenuse facing the top-left.

- **Step 4: Install PHP**

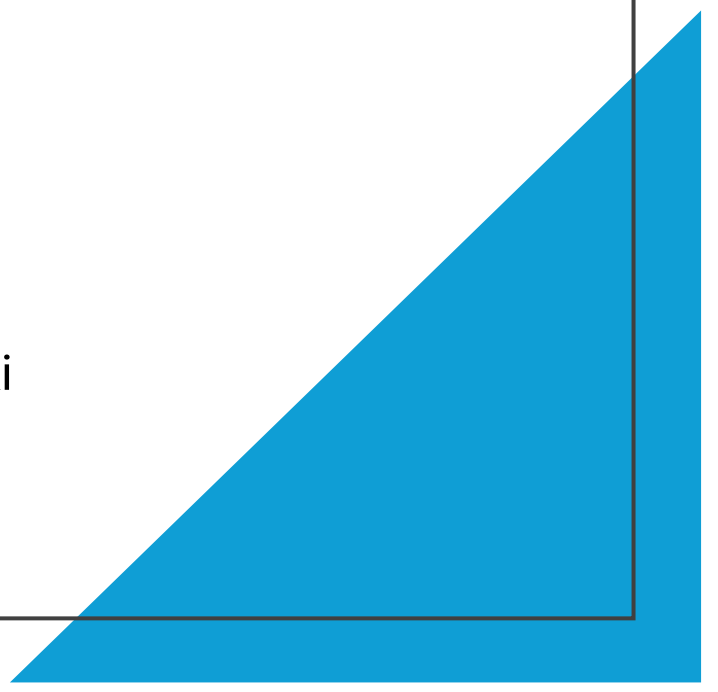
- `sudo apt install php libapache2-mod-php php-mysql php-xml php-mbstring php-intl php-apcu php-json php-curl -y`
- `sudo systemctl restart apache2`



## Step 5: Download and Configure MediaWiki

- `cd /var/www/html`  
`sudo wget`
- <https://releases.wikimedia.org/mediawiki/1.41/mediawiki-1.41.1.tar.gz>
- `sudo tar -xvzf mediawiki-1.41.1.tar.gz`
- `sudo mv mediawiki-1.41.1 mediawiki`
- `sudo chown -R www-data:www-data /var/www/html/mediawiki`

Access installer: <http://mediawiki>

A large blue right-angled triangle is positioned in the bottom right corner of the slide, pointing towards the top-left.

## **Step 6: Web Installer Configuration**

- Choose language
- Verify environment
- Connect to MySQL database using credentials above
- Configure wiki name, admin username and password
- Download LocalSettings.php and upload to /var/www/html/mediawiki



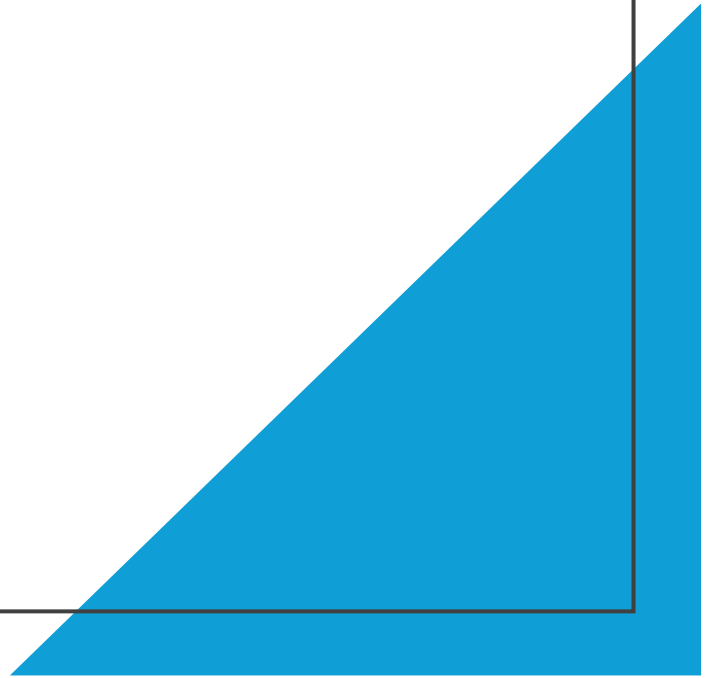
# Backup Strategy for MediaWiki Server

- A **backup mechanism** ensures that in the event of hardware failure, software corruption, or accidental deletion, **MediaWiki content** can be fully restored. MediaWiki stores content in two main locations:
-

## **2 locations -**

protect both, we implement a **comprehensive, automated daily backup** that includes:

- MySQL database (wikidb)
- Uploaded images folder (/images)
- Main configuration file (LocalSettings.php)





# Backup Components

| Component        | Location                                  | Type         |
|------------------|-------------------------------------------|--------------|
| MySQL Database   | mediawiki                                 | Logical dump |
| Uploaded Files   | /var/www/html/mediawiki/images/           | Static files |
| Configuration    | /var/www/html/mediawiki/LocalSettings.php | Config file  |
| Extensions/Skins | /var/www/html/mediawiki/extensions/       | Source files |

# Manual Backup Script (backup\_mediawiki.sh)

```
#!/bin/bash

# Configurable Variables
BACKUP_DIR="/var/backups/mediawiki"
DATE=$(date +%F_%H-%M)
DB_NAME="mediawiki"
DB_USER="root"
DB_PASS="your_mysql_password"

# Ensure backup directory exists
mkdir -p "$BACKUP_DIR"

echo " Starting MediaWiki backup @ $DATE..."

# 1. Backup MySQL database
mysqldump -u "$DB_USER" -p"$DB_PASS" "$DB_NAME" > "$BACKUP_DIR/db_$DATE.sql"
```

# 2. Backup uploaded files

```
tar -czf "$BACKUP_DIR/images_$(date +%Y%m%d).tar.gz" /var/www/html/mediawiki/images
```

# 3. Backup configuration and extensions

```
tar -czf "$BACKUP_DIR/config_$(date +%Y%m%d).tar.gz" \  
/var/www/html/mediawiki/LocalSettings.php \  
/var/www/html/mediawiki/extensions \  
/var/www/html/mediawiki/skins
```

# 4. Log

```
echo " Backup completed: $BACKUP_DIR"
```

# Schedule Cron Job

## **Edit the cron table:**

`crontab -e`

- Add this line to run the backup daily at 2:00 AM:
- `0 2 * * * /usr/local/bin/backup_mediawiki.sh`
- All backups will be stored under `/var/backups/mediawiki`

## **To Verify:**

Check cron logs:

`bash`

`grep CRON /var/log/syslog`

Or confirm file timestamps in `/var/backups/mediawiki/`

# Restore Steps (Disaster Recovery)

## A. Restore Database:

```
mysql -u root -p mediawiki < db_2025-07-31_02-00.sql
```

## B. Restore Files:

```
tar -xzf images_2025-07-31.tar.gz -C  
/var/www/html/mediawiki/images  
tar -xzf config_2025-07-31.tar.gz -C  
/var/www/html/mediawiki/
```

Restart services:

```
sudo systemctl restart apache2 mysql
```



# Fallback Plan

## **Why Linux Setup Failed (Example Scenarios)**

You can customize this based on what you encountered. Example reasons:

- VirtualBox guest additions failed, preventing proper display resolution.
  - MySQL service failed to start due to missing dependencies.
  - MediaWiki version not compatible with installed PHP version.
  - Firewall or NAT issues blocked apt or wget from retrieving packages.
-

# Tools used for Fallback Deployment

| Tool                   | Description                                      |
|------------------------|--------------------------------------------------|
| WAMP Server            | All-in-one stack: Windows + Apache + MySQL + PHP |
| MediaWiki (.zip)       | Latest stable release from mediawiki.org         |
| phpMyAdmin             | MySQL GUI for creating DB and managing access    |
| Windows Task Scheduler | For automated backup scripts                     |
| Notepad++              | For editing config files (LocalSettings.php)     |

# Verification Checklist

| Component           | Command / URL                     | Expected Result                         |
|---------------------|-----------------------------------|-----------------------------------------|
| Apache              | systemctl status apache2          | Service status: <b>Active (running)</b> |
| MySQL               | sudo mysql -u root -p             | Opens <b>MySQL shell</b> without error  |
| PHP                 | php -v                            | Displays <b>installed PHP version</b>   |
| phpMyAdmin          | http://localhost/phpmyadmin       | <b>Login page</b> loads in browser      |
| PHP Info            | http://localhost/info.php         | Displays <b>PHP configuration page</b>  |
| MediaWiki           | http://localhost/mediawiki        | Loads <b>MediaWiki homepage</b>         |
| Backup Script       | Run .sh or .bat manually          | SQL and file <b>backups created</b>     |
| Cron/Task Scheduler | crontab -l / check Task Scheduler | <b>Scheduled job listed correctly</b>   |



# Mistakes Face and Troubleshooting

## 1) MySQL Authentication Plugin

- Faced login issue with wikiuser.

## 2) Database storage issues

## 3) Extensions Missing

- Initially missed php, Apache
- Setup page showed a warning.
- Fixed by installing: `sudo apt install php-intl`

## 4) Apache permission Error

- Got Forbidden Error
  - Resolved by setting correct permissions and Allow Override All.
-

# Installation Troubleshooting

| Problem                                     | Cause                                                                           | Solution                                                                                                                                                          |
|---------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Apache not starting</b>                  | Port 80 already in use                                                          | Run: <code>sudo lsof -i :80</code> → kill or reconfigure the conflicting service                                                                                  |
| <b>MediaWiki not loading in browser</b>     | 1. Apache not running<br>2. Wrong DocumentRoot<br>3. Incorrect file permissions | Check with:<br><code>sudo systemctl status apache2</code><br><code>ls -l /var/www/html/</code><br>Use <code>sudo chown -R www-data:www-data /var/www/html/</code> |
| <b>PHP pages download instead of render</b> | PHP module missing in Apache                                                    | Run: <code>sudo apt install libapache2-mod-php</code><br>Then restart: <code>sudo systemctl restart apache2</code>                                                |
| <b>MySQL/MariaDB Access Denied</b>          | Wrong password or root login disabled                                           | Use: <code>sudo mysql -u root</code><br>If needed: <code>sudo mysql_secure_installation</code> to reset root auth                                                 |
| <b>Can't access phpMyAdmin</b>              | phpMyAdmin not enabled                                                          | Run: <code>sudo ln -s /usr/share/phpmyadmin /var/www/html/phpmyadmin</code><br>Ensure Apache is restarted                                                         |

# Extra Features Implemented

- Daily cron-based backup
- phpMyAdmin for database management
- MediaWiki extensions for visual editing
- Enforced HTTPS using self-signed certificate
- Added user accounts for group-based wiki editing

# Optional Extras part

- Add MediaWiki extensions:
  - e.g., VisualEditor or SyntaxHighlight
  - Screenshot: Installed Extensions in MediaWiki
  - Enable HTTPS using Let's Encrypt and capture:
  - `bash`
  - `sudo apt install certbot python3-certbot-apache sudo certbot --apache`
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# Conclusion

- MediaWiki successfully installed on LAMP stack. With all requirements.
- Backup system created for daily database and file archival.
- final setup that ensures both the task.

THANK YOU

